

WINDER BORROW, GA, USA

WMO: 747809

Lat: 33.983N

Lon: 83.668W

Elev: 943

StdP: 14.20

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 03832

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.6	28.0	0.8	5.9	38.0	8.6	8.9	35.6	23.1	40.0	20.2	40.3	5.9	320	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	95.1	72.8	92.7	73.0	90.6	72.9	78.3	88.2	77.3	87.4	76.4	85.9	5.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	137.8	81.7	73.9	130.7	80.5	73.2	127.7	80.0	42.3	88.5	41.4	86.6	40.4	86.6	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.2	15.7	12.6	DB	18.1	98.0	2.9	3.5	16.0	100.6	14.3	102.6	12.7	104.6	10.6	107.2
			WB	15.9	80.2	2.6	3.2	14.0	82.5	12.4	84.3	10.9	86.1	9.0	88.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	43.7	46.9	55.1	62.3	70.2	77.8	80.2	79.5	74.3	63.6	53.7	47.1
	DBStd	14.62	9.15	8.33	8.41	7.11	6.23	4.39	3.87	3.63	5.28	7.48	7.68	8.04
	HDD50	642	238	145	46	6	0	0	0	0	0	5	46	156
	HDD65	2672	659	509	322	131	29	1	0	0	4	113	348	556
	CDD50	5371	44	59	202	375	627	833	936	915	730	426	157	67
	CDD65	1927	0	2	14	50	191	384	471	450	284	70	9	2
	CDH74	17568	0	10	133	529	1655	3644	4639	4141	2267	505	43	2
	CDH80	7085	0	0	12	96	528	1569	2116	1813	844	105	2	0
(14) Wind	WSAvg	4.7	6.1	6.0	6.1	5.5	4.5	3.8	3.5	2.9	3.5	4.3	4.7	5.4
(15) Precipitation	PrecAvg	52.1	4.9	4.5	5.7	3.5	3.7	4.8	4.8	4.3	4.3	3.1	4.1	3.8
	PrecMax	63.9	8.3	8.4	9.5	9.4	9.1	10.2	11.3	8.7	12.2	8.4	8.3	8.3
	PrecMin	40.0	2.5	1.6	1.2	1.6	1.4	1.3	1.4	1.4	0.0	0.3	1.0	2.1
	PrecStd	6.4	1.5	1.8	2.2	1.9	1.7	2.3	2.7	2.1	3.0	2.2	2.0	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	74.9	81.6	86.2	91.1	96.8	97.2	97.4	94.9	89.6	79.1	72.7
		MCWB	60.1	62.7	60.0	65.1	67.1	72.9	73.5	72.9	70.2	71.3	64.4	62.6
	2%	DB	64.5	69.6	78.6	82.1	88.3	93.2	94.9	94.6	90.8	82.1	73.3	68.5
		MCWB	58.6	58.2	60.1	62.9	68.5	72.6	73.5	73.0	70.2	67.4	61.7	62.0
	5%	DB	62.8	64.3	73.2	80.7	86.1	90.7	91.7	91.1	88.3	79.4	70.4	64.1
		MCWB	56.6	56.0	58.6	62.3	67.8	72.0	74.2	73.8	70.8	65.2	60.5	58.1
	10%	DB	58.7	61.4	70.2	77.1	82.1	88.4	90.3	89.7	85.7	76.7	67.7	61.1
		MCWB	52.8	52.3	57.2	60.7	67.2	72.1	74.3	73.5	69.8	64.8	58.5	55.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.3	66.1	66.5	70.0	75.1	79.4	80.0	79.4	77.3	75.0	71.0	67.1
		MCDB	66.6	71.3	73.0	76.8	83.6	89.4	89.8	88.6	83.9	82.1	73.9	70.3
	2%	WB	61.3	62.7	64.1	67.7	73.2	77.4	78.2	78.0	75.4	72.8	67.0	63.4
		MCDB	63.5	66.5	70.4	75.7	81.8	88.0	88.4	88.1	82.9	78.8	70.9	66.0
	5%	WB	58.1	58.9	62.0	65.8	71.8	76.1	77.3	77.1	74.2	70.6	63.4	60.8
		MCDB	61.2	62.6	68.4	74.7	80.3	85.7	87.3	87.3	81.7	75.4	66.8	63.9
	10%	WB	54.2	55.0	59.9	63.7	70.2	74.8	76.4	76.2	73.0	68.1	60.6	56.7
		MCDB	56.9	59.8	67.3	71.8	77.6	83.7	86.0	85.6	80.4	72.7	65.0	60.8
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	18.5	20.9	22.0	20.3	19.1	18.3	17.9	18.5	19.7	20.7	17.6
		MCDBR	21.3	22.3	24.8	26.1	23.3	22.0	20.5	21.3	22.2	22.8	23.7	20.6
	5% WB	MCWBR	15.2	14.2	13.0	11.4	8.7	7.5	7.3	7.3	7.9	10.2	13.4	14.5
		MCWBR	17.2	16.9	20.2	21.1	18.5	18.4	18.3	18.0	16.4	16.6	18.1	17.7
(40) Clear-Sky Solar Irradiance	taub		0.311	0.316	0.351	0.388	0.447	0.476	0.516	0.514	0.418	0.376	0.339	0.309
		taud	2.524	2.503	2.432	2.315	2.189	2.151	2.059	2.050	2.303	2.407	2.460	2.550
	Ebn at Noon		287	297	293	285	268	259	248	246	267	271	273	281
		Edh at Noon	26	30	35	41	47	49	53	52	39	32	28	24
(44) All-Sky Solar Radiation	RadAvg		858	1042	1384	1732	1917	1981	1882	1733	1503	1266	952	729
	RadStd		50	128	101	114	151	153	131	93	147	149	97	75

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+301	+200

Nomenclature: See separate page